APPSC INDUSTRIAL PROMOTION OFFICER

Complete Preparation Guide — Paper I & Paper II

Purpose: A compact revision and practice guide. Use the official APPSC notification/syllabus as the final authority for the exact scheme and topics.

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1. EXAM-ORIENTED PREPARATION STRATEGY

• **Step 1 — Master the syllabus:** Divide every syllabus heading into small subtopics and mark each as Strong / Average / Weak.

• **Step 2 — Build Paper II first:** Give extra attention to Industrial Engineering, Industrial Management, Production, Materials, Safety, Quality, Entrepreneurship and basic Science/Engineering.

• **Step 3 — Daily MCQs:** Practise 50–100 questions and maintain an error notebook.

• **Step 4 — Current affairs:** Revise recent national, Andhra Pradesh, economy, industry, science & technology developments.

• **Step 5 — Mock tests:** Practise with a timer and analyse every incorrect answer.

• **Negative marking:** Follow the exact marking rule in the current official notification; avoid blind guessing.

2. PAPER I — GENERAL STUDIES & MENTAL ABILITY

Indian Polity & Governance

• Constitution: Preamble, Fundamental Rights, DPSP, Fundamental Duties.

• Union and State executive: President, Governor, Prime Minister, Chief Minister and Councils of Ministers.

• Parliament and State Legislature; constitutional and statutory bodies.

• Local government: Panchayati Raj and urban local bodies.

• Judiciary, elections, public administration, transparency and governance.

Indian History & Andhra Pradesh

• Ancient and medieval Indian history: major dynasties, culture and architecture.

• Modern India: British rule, reform movements, freedom struggle and important Acts.

• Post-independence India and major developments.

• Andhra Pradesh: historical movements, formation of the State, culture, economy and important personalities.

Geography & Environment

• Physical geography: landforms, climate, rivers, soils and natural resources.

• Indian agriculture, industries, transport and population.

• Andhra Pradesh geography: rivers, agriculture, industries, ports and resources.

• Ecosystems, biodiversity, pollution, climate change, conservation and sustainable development.

Indian Economy & Industry

• National income, inflation, unemployment, fiscal and monetary policy.

• Banking, taxation, public finance and basic economic indicators.

• Industrial development, MSMEs, infrastructure, investment and entrepreneurship.

• Government schemes connected with employment, industry and economic development.

General Science & Technology

• Basic physics, chemistry, biology and everyday science.

• Information technology, communication, biotechnology and emerging technologies.

• Energy, renewable energy, health technology and environmental technology.

Mental Ability

• Number series, analogies, coding-decoding, syllogisms and classifications.

• Percentages, ratios, averages, profit and loss, time and work, time-distance.

• Data interpretation: tables, charts, percentages and basic statistics.

• Logical reasoning, statements, assumptions, directions and simple puzzles.

3. PAPER II — APPLIED SCIENCE, ENGINEERING & MANAGEMENT

Engineering Mathematics

• Algebra: equations, logarithms, progressions and basic matrices.

• Trigonometry: identities, heights/distances and basic applications.

• Differential and integral calculus: standard derivatives/integrals and applications.

• Statistics and probability: mean, median, mode, dispersion and elementary probability.

Physics

• Units and dimensions; motion, force, work, power and energy.

• Heat, temperature, thermodynamics and basic fluid concepts.

• Electricity: current, voltage, resistance, power, series/parallel circuits.

• Magnetism, electromagnetic induction, optics, sound and basic semiconductor concepts.

Chemistry

• Atomic structure, periodic table and chemical bonding.

• Acids, bases, salts, pH and common industrial chemicals.

• Metals, alloys, corrosion and prevention.

• Organic chemistry basics, polymers, fuels, water treatment and industrial chemistry.

Industrial Engineering

• Work study: method study and work measurement.

• Time study, standard time, productivity and efficiency.

• Plant layout, material handling and facility planning.

• Operations research basics: linear programming, inventory and queuing concepts.

Industrial Management

• Functions of management: planning, organizing, staffing, directing and controlling.

• Leadership, motivation, communication and decision-making.

• Organization structures and coordination.

• Productivity, cost control, maintenance and project basics.

Production & Operations Management

• Production systems: job, batch, mass and continuous production.

• Production planning and control; routing, scheduling, dispatching and follow-up.

• Forecasting, capacity planning and line balancing.

• Lean concepts, waste reduction, 5S and continuous improvement.

Materials Management

• Purchasing, stores management and inventory control.

• EOQ, reorder level, safety stock and ABC analysis.

• Vendor selection, material handling, codification and standardization.

• Supply chain basics and logistics.

Quality Management

• Quality concepts, inspection and quality assurance.

• PDCA, continuous improvement and basic statistical quality control.

• Control charts and process capability concepts.

• TQM, quality circles, customer focus and ISO 9001 fundamentals.

Industrial Safety

• Hazard identification, risk assessment and safe work practices.

• Personal protective equipment, machine guarding and electrical safety.

• Fire safety, emergency response and accident prevention.

• Occupational health, safety training and safety management.

Entrepreneurship

• Entrepreneur characteristics and functions.

• Idea generation, feasibility study and business plan.

• Sources of finance, MSME concepts and basic project appraisal.

• Marketing, risk management and government support for enterprises.

Computer Fundamentals

• Computer hardware, software, operating systems and networking basics.

• Word processing, spreadsheets, presentations and databases.

• Internet, email, cybersecurity, data privacy and common digital services.

4. FORMULA & QUICK-REVISION SHEET

Topic	Key formula / point
Percentage	Percentage = (part / whole) × 100
Average	Average = Sum of observations / Number of observations
Simple Interest	SI = P × R × T / 100
Compound Amount	A = P(1 + R/100)^n
Speed	Speed = Distance / Time
Work	If a worker completes work in T days, rate = 1/T per day
Ohm's Law	V = IR
Electrical Power	P = VI = I^2R = V^2/R
Kinetic Energy	KE = ½mv^2
Potential Energy	PE = mgh
Density	Density = Mass / Volume
Pressure	Pressure = Force / Area
Efficiency	Efficiency = Output / Input × 100
Productivity	Productivity = Output / Input
EOQ	EOQ = √(2DS/H), where D=demand, S=ordering cost, H=holding cost
Break-even	Break-even point occurs where total cost = total revenue

5. IMPORTANT PRACTICE MCQs

1. Which principle states that a process should be improved continuously?

A) Continuous improvement / Kaizen B) Depreciation C) Outsourcing D) Forecasting

Answer: A) Continuous improvement / Kaizen

2. In Ohm's law, V represents:

A) Power B) Voltage C) Resistance D) Current

Answer: B) Voltage

3. ABC analysis is mainly used for:

A) Recruitment B) Quality inspection C) Inventory control D) Plant safety

Answer: C) Inventory control

4. EOQ is primarily concerned with:

A) Economic order quantity B) Employee qualification C) Equipment output D) Energy quality

Answer: A) Economic order quantity

5. PDCA stands for:

A) Plan–Develop–Control–Assess B) Produce–Do–Check–Act C) Plan–Design–Calculate–Act D) Plan–Do–Check–Act

Answer: D) Plan–Do–Check–Act

6. A chart used to study causes of a problem is commonly called:

A) Gantt chart B) Fishbone diagram C) Pie chart D) Histogram

Answer: B) Fishbone diagram

7. The main purpose of PPE is to:

A) Increase production speed B) Replace training C) Reduce worker exposure to hazards D) Eliminate all machinery

Answer: C) Reduce worker exposure to hazards

8. In a series electrical circuit, the current is:

A) The same through each element B) Always zero C) Different at every point D) Independent of resistance

Answer: A) The same through each element

9. TQM primarily emphasizes:

A) Final inspection only B) Organization-wide quality and customer focus C) Reducing training D) Increasing inventory

Answer: B) Organization-wide quality and customer focus

10. The Constitution of India came into force on:

A) 15 August 1947 B) 26 November 1949 C) 26 January 1950 D) 2 October 1950

Answer: C) 26 January 1950

11. The Fundamental Duties are contained in:

A) Article 51A B) Article 32 C) Article 21 D) Article 14

Answer: A) Article 51A

12. Inflation generally refers to:

A) Fall in employment only B) A sustained rise in the general price level C) Increase in exports only D) Fall in money supply only

Answer: B) A sustained rise in the general price level

13. A Gantt chart is mainly used for:

A) Chemical analysis B) Inventory valuation only C) Electrical testing D) Scheduling and tracking activities

Answer: D) Scheduling and tracking activities

14. 5S is associated with:

A) Workplace organization B) Monetary policy C) Tax collection D) Product advertising

Answer: A) Workplace organization

15. In a break-even situation:

A) Profit is maximum B) Variable cost is zero C) Total revenue equals total cost D) Sales are zero

Answer: C) Total revenue equals total cost

6. 30-DAY STUDY PLAN

Days	Focus
1–3	Official syllabus + Polity fundamentals + current affairs
4–6	History, Andhra Pradesh history and geography
7–9	Economy, industry, environment and science
10–12	Mental ability, arithmetic and data interpretation
13–15	Engineering Mathematics + Physics
16–18	Chemistry + Computer Fundamentals
19–21	Industrial Engineering + Work Study
22–23	Industrial Management + Production Management
24–25	Materials Management + Inventory
26	Quality Management + TQM + ISO basics
27	Industrial Safety + Entrepreneurship
28	Full Paper I mock + error analysis
29	Full Paper II mock + error analysis
30	Final revision: formulas, weak areas and current affairs

FINAL REVISION CHECKLIST

✓ Read the official syllabus and notification before relying on any coaching material.

✓ Complete at least 3 revision cycles for Paper II.

✓ Maintain a notebook of wrong MCQs and revise it every 2–3 days.

✓ Practise calculations without a calculator where the exam rules require it.

✓ Use official APPSC updates for dates, scheme, eligibility and final syllabus.